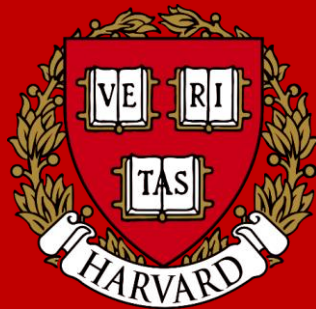


HARVARD UNIVERSITY

MGMT 5515 - THE BUSINESS OF FOOTBALL: THE WORLD'S SPORT

“THE BUSINESS OF LIVERPOOL FOOTBALL CLUB”



NITESH CHAWDA

HUID: 20923245

NITESHCHAWDA@G.HARVARD.EDU

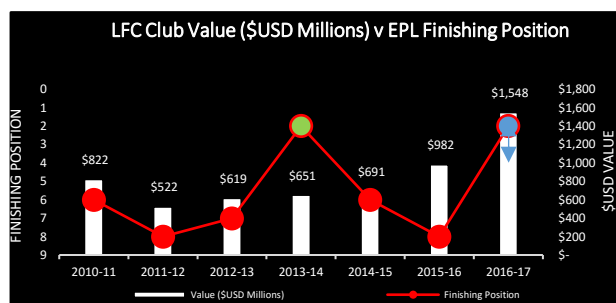
CONTENTS

Section 1: Introduction	Page 1
Section 2: Assessing Liverpool as Corporation.....	Page 1-4
Section 3A: Liverpool as an Organic Organization – Picking the Team.....	Page 4-6
Section 3B: Liverpool as an Organic Organization – The Workforce (Players)	Page 6-8
Section 5: Conclusion and Recommendations.....	Page 9
Section 6: Appendices.....	Page 10

1. INTRODUCTION: IF MONEY HASN'T LEAD TO SILVERWARE, WHAT IS LFC MISSING?

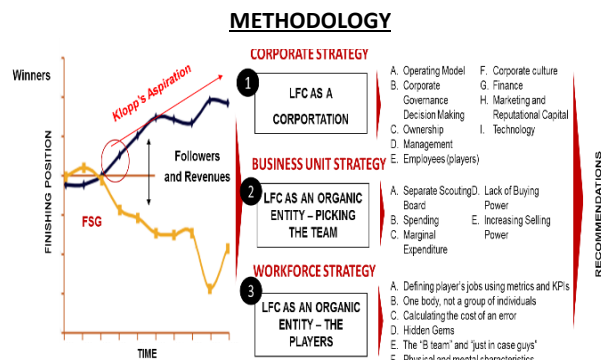
A. IS SYMANKSI WRONG? DESPITE INCREASING SPEND, LFC HAVE STAGNATED FOR ~27 YEARS

Szymanski claims that “expenditure is a very reliable indicator of a team’s success”¹. Since Fenway Sport Group’s (“FSG”) acquisition of Liverpool Football Club (“LFC”) the club’s net value has increased (66%) from \$522 million in (2011-12)² to \$1,548 million (2016-17), making LFC the 8th most valuable club globally.³ However as seen below, LFC continue to stagnate as a mid-table team.



B. IS THERE A CASE FOR CHANGE UNDER FSG?

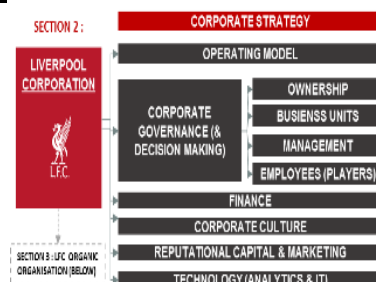
In 2010, John Henry (FSG group) successfully purchased LFC. This signalled more than just a clearing of debts and revenue, but signalled the beginning of a “new era” as Henry is globally renowned as the guru of turning underperforming and undervalued teams into champions (*see footnote*⁴ demonstrating Henry’s success famously turning Oakland As and Boston Red Sox into champions). This paper analysis the strategic and economic position of **1. LFC a Corporation** and **2 LFC as an Organic Organisation** (football club).



2. LIVERPOOL AS A CORPORATION

A. THE FALL FROM EU CHAMPIONS TO A TEAM DESPERATE FOR CHANGE

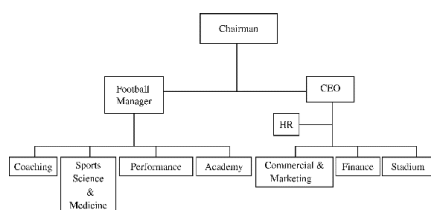
It has been 27 years, 20 hours, 42 minutes and 4 seconds; 4.5 years before Raheem Sterling was to be born; and George H.W Bush was President negotiating with the Soviet Union, since LFC has won the English League.⁵ Fast forward to today, Liverpool has been displaced among from the “handful EPL teams that dominate, athletically and in every other way”⁶, losing confidence and reputation. FSG promised “another way of winning”⁷, starting with the corporate value levers depicted to the right.



B. OPERATING MODEL

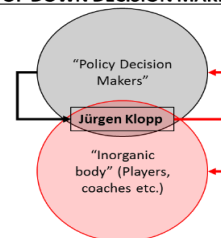
FSG’s first order of business was to impose a clear corporate operating model (“OM”). Due to the stark contradiction between football (based on biological and subjective factors), this objective and institutionalised approach contradicted the intuitive understanding of football. The OM separates on field tactics and has brought about stability, via “a functional structure so that each department grouped according to its purpose; marketing, sales, production etc.”⁸ Albeit well defined design principles, FSG has deliberately “lowered boundaries, allowing strategic coordination and communication between departments.”⁹ This purposely embodies Bill Shankly’s 1969 philosophy that “At LFC, there's a holy trinity – the players, the manager and the supporters (where) Directors do not come into it and are (strictly) only there to (run the business) .”¹⁰ This is depicted below:¹¹

LFC OPERATING MODEL



Chairman: Tom Werner, Manager: Jurgen Klopp, CEO : Peter Moore
This structure, can be traced back to when the Oakland A's imposed what was known as the "executive team" composed of objective management staff and statisticians at the club level and a divisional committee to represent players and "middle managers"

TOP DOWN DECISION MAKING



¹ Money and Soccer, A socceroconomics Guide, Stefan Szymanski Chapters 4 on Revenue and chapter 5 on Debt

² Note: The decrease from \$822 to \$522 from 2010 to 2011 was due to FSG agreeing to pay out \$300 million of Liverpool debts upon acquisition, www.thisisanfield.com/2014/03/fsg-reduced-lfc-debts-200m/, 01/28/2017

³ The Business of Soccer: Forbes, <http://www.forbes.com/teams/liverpool/>, 01/28/2017.

⁴ John Henry took much of his learnings from William Daro, “Billy Beane”, previous GM of the Oakland as (winning 102 games from spending only \$34m)⁴ & John Henry failed recruit Beane as GM of the Boston Red Sox in 2002.

⁵ 1:27 PM Melbourne Time, 6th May 2017, <http://since.liverpoolfc.com/league-title/>

⁶ Money and Soccer, A socceroconomics Guide, Stefan Szymanski Chapters 1 on Dominance and Distress

⁷ <http://anfieldindex.com/6265/fsg-liverpool-fc-another-way-winning.html>

⁸ <http://businessstudiesleadershipemerson.blogspot.com.au/2013/09/what-is-liverpools-organizational.html>

⁹ Ibid.

¹⁰ <https://www.theguardian.com/football/2009/oct/18/bill-shankly-liverpool-manager>

¹¹ <http://businessstudiesleadershipemerson.blogspot.com.au/>



C. GOVERNANCE & DECISION MAKING

As depicted, LFC has a clear governance structure with decision making clearly delineated into 1. A central policy makers (e.g. executive committee, transfer committee etc.) and 2. The field/team decision makers. Klopp has a significant role in the former and is the head of the later in charge of all sub-ordinate managerial roles in the “inorganic” field team. Thus, as seen in the Venn diagram, Klopp is central to the entity and all other “middle managers” must subordinate individual decisions and interests to policy makers and play the role of implementing policies (aka coaching and training players). However, decision makers are bound to criteria which “uses a unique objective approach to all decisions”,¹² combining sabermetrics (see footnote¹³ detailing sabermetrics) to “expose the illusions”¹⁴ behind subjective decisions and “disapprove traditional pure wisdom as a guiding principles (and) thereby prejudices”.¹⁵ Furthermore, FSG has extended “win maximisation”¹⁶ to decision making that concentrates on economies of scale, focusing on the “marginal commitment to attain the next player’s individual target”¹⁷ (see section 4A that elaborates on this point).

D. OWNERSHIP

As Szymanski highlights FSG (owners) “derives a benefit only if LFC wins, and to win LFC relies on Henry as a sugar daddy who will fund the enterprise...buy teams as a status symbols, seeking to share in the glory of championships”.¹⁸ The below however highlights that Henry is not merely a sugar daddy nor a complete “fandom”¹⁹ but a passionate sports owner who wishes to replicate his experience with the Boston Red Sox at Liverpool.²⁰

5 reasons why John F Henry Purchased Liverpool:

- Similarities between Liverpool and Boston:** “Both cities will exorcise me for comparing them, but like it or not they are kindred spirits.”
- Liverpool and the Red Sox have the best rivalries in their sport:** “When we play our rival Manchester United the audience on television is about a billion people. That’s about nine times the size of the audience for the Super Bowl. This is very similar to the relationship between the Red Sox and the Yankees.”
- Both teams have the most exciting player in the world:** “Liverpool have a David Ortiz. His name is Luis Suarez. He leads the league in goals and he’s the most exciting player in the world today.”
- Both clubs have iconic stadiums:** They play in the most historical cathedrals of sport. Anfield has 7,000 more seats than Fenway [the Red Sox’s stadium] but actually you might say it’s a bit cosier than Fenway.”
- But John Henry’s spending sprees haven’t always paid off:** “When the Red Sox spent \$300million on Adrian Gonzalez and Carl Crawford, Liverpool fans were irate. It actually should have been Boston fans. “When Liverpool spent \$200m on soccer’s players [Henderson, Downing, Andy, Suarez and others] the same year, not including their salaries, Red Sox fans were irate. But it should have been Liverpool fans.”

However, as highlighted above, Henry remains distant from on the field play and acts as a “corporate boss” corporation while Klopp runs “the show and every action of the player(s)”.²¹ This is due to Henry’s “beat” to sports²² and treats the game as a “market that intrinsically tells a story...more than what you see on the pitch”.²³ As such Henry has removed the traditional mentality that the manager is responsible for all pressure, and implemented a RACI²⁴ framework that means even executives must equally share responsibilities. This recently seen when Ian Ayre (previous CEO) retired, due to “Simply not being able deal with the 24/7 demands of (FSG and) the job any longer”²⁵ and Henry re-iterated by Bill Shankly’s philosophy that ‘nobody is greater than the interest of the club’ and FSG put “club ahead of Ayer’s own interest”.²⁶ Binding all to the same standards has aligned LFC entirely meaning that “heads (at all levels) can roll very far for incorrect decisions”.²⁷

E. MANAGEMENT

Following to the Henry’s division of accountability, LFC has worked publicly with Szymanski to remove the image of Klopp as the “Messiah, a man of extraordinary capabilities who can turn straw into gold (and has allowed Klopp to focus) on outcomes and not processes.”²⁸ FSG has encouraged Klopp to shift from “being like ‘a GM’, drilling players”²⁹ to

¹² Money ball, Michael Lewis, Chapter 2 on “How to find a ballplayer”.

¹³ **Sabermetrics** is the empirical analysis of sport, especially baseball statistics that measure in-game activity. Sabermetricians collect and summarize the relevant data from this in-game activity to answer specific questions.

¹⁴ Ibid at 12.

¹⁵ Money ball, Michael Lewis, Chapter 3 “The Enlightenment”.

¹⁶ Money and Soccer, A socceroconomics Guide, Stefan Szymanski, Chapter 6 on “Ownership”.

¹⁷ Money ball.

¹⁸ Ibid at 16.

¹⁹ Harvard Extension School, MGMT 5515, Dale Sheptak.

²⁰ <http://www.mirror.co.uk/sport/football/news/john-w-henry-buy-liverpool-3020975>.

²¹ Money ball, Michael Lewis, Chapter 7 on “Giambi’s Hole”.

²² Money ball, Michael Lewis, Page 91.

²³ Ibid.

²⁴ RACI: Ensures decisions and people are responsible, accountable, consulted and informed

²⁵ <http://www.mirror.co.uk/sport/football/news/revealed-ian-ayre-quitting-liverpool-7559153>

²⁶ Ibid.

²⁷ Money ball, Michael Lewis, Afterword.

²⁸ Money and Soccer, A socceroconomics Guide, Stefan Szymanski, Chapter 7 on “Strategy”/

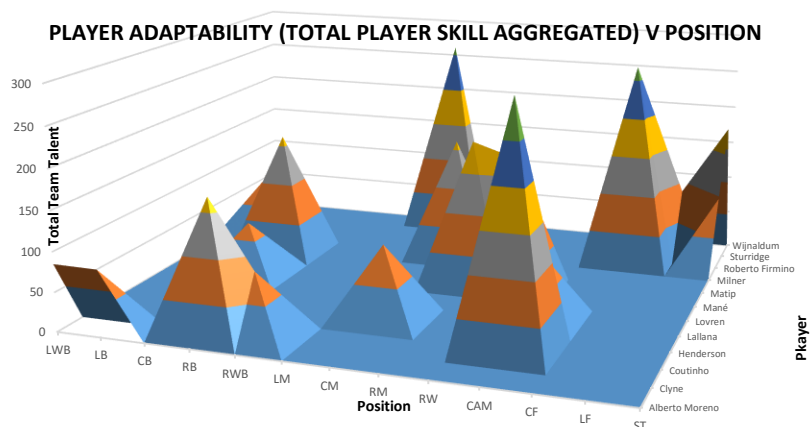
²⁹ Ibid at 21.

focusing on being equal with the players and as famously quoted to act like the “normal one”.³⁰ Likewise, Klopp notably re-aligned expectations to remove the “psychologically scarring forged from the failure to win the title for 27 years”.³¹ This has been done by directly addressing fans that “he is not the messiah and that Liverpool will likely not win every game 8-0 under his charge”³² and that even “missed goals” or “draws / losses” should not bemoaned upon or booed at, but be celebrated as opportunities “in the moment that should be enjoyed”³³. This view that to do something right, there must be errors (see **Appendix 1: Dialectics**) is empowered by FSG’s ‘sabermetrics and Opta stats’ helps identify “errors as false positives”³⁴ (see **footnote**³⁵: Examples - Boston Red Sox). However, inevitably “football is not mistakes of men v men, it’s men v circumstance”³⁶ and previous managers over dependent on the data as the “strategy that was statistically least likely to fail and but not the most effective”³⁷ eventually, among other reasons leading to their downfall (e.g. Brendan Rodgers³⁸).

F. EMPLOYEES (AKA PLAYERS)

FSG endorses that LFC is not about individual units and players are part of a normal workforce, with flat and cohesive organisational structure and equal responsibility”.³⁹ Therefore, players are incentivised by Klopp for meeting expectations and disciplined if not. This again embodies Bill Shankly’s ethos that “No one is bigger than the club” and that it is a “earned privilege to play for LFC”⁴⁰ (see **footnote**⁴¹ on Mamadou Sakho being made redundant for missing a meeting). The purpose of this is to encourage players to adopt “set routines and deviate old player routines (and to remove) individualised and non-team functionally aligned tactics”.⁴² This is accepted as a “natural internal ecosystem”⁴³ that must exist in order for Klopp to apply his style of play and with new managers (similar to the workforce) player purchased for predeceasing tactics are made redundant (see **footnote**⁴⁴ listing players made redundant due to an inability to Brendan Rodgers’ Tiki-taka Klopp’s high pace fluid football).

Furthermore, Klopp expects employees to be multi-functional and flexible to adopt various positions, which inherently is an unconventional approach “to stand against the system”.⁴⁵ The below depicts the fluidity of the current LFC squad with players being able to play in multiple roles maintaining an average of with 6 of the players being able to perform more than one role.



G. CORPORATE CULTURE

In addition to the culture that the “manager does not come first”⁴⁶ the LFC culture is guided by comradery and players learning to work together “by smell and (game play)”.⁴⁷ Furthermore, a culture of continuously learning is encouraged and players are expected to approach the game with “how to play pro football”⁴⁸ enabling players to be aware of what is happening on the pitch and what Klopp expects looking outside in. This gives a cohesive view and ensures the body is playing towards the common strategy (see **Appendix 2: Sturridge circulating paper notes on the pitch**).



³⁰ <http://www.telegraph.co.uk/sport/football/managers/jurgen-klopp/12095183/Liverpool-apply-to-trademark-The-Normal-One-in-honour-of-Jurgen-Klopp.html>.

³¹ <http://www.telegraph.co.uk/football/2017/01/20/liverpool-manager-jurgen-klopp-issues-rallying-cry-supporters/>

³² <https://www.balls.ie/football/jurgen-klopp-offers-blunt-message-to-optimistic-liverpool-fans-313575>

³³ <http://www.telegraph.co.uk/football/2017/01/20/liverpool-manager-jurgen-klopp-issues-rallying-cry-supporters/>

³⁴ Money ball, Michael Lewis, Chapter 4 on “Field of Ignorance”.

³⁵ Boston Red Sox have used Sabermetrics to think “two steps ahead”. For instance, instead of looking at the hitter, you cannot tell if a fielder is useless unless the ball gets to them and such have no visibility, thus using statistics and situational testing, range factors were introduced to highlight “2 steps before”, too look at the fielder instead of the hitter highlighting missed things.

³⁶ Ibid at 34.

³⁷ Ibid at 34.

³⁸ <http://bleacherreport.com/articles/2296954-inside-line-liverpools-transfer-committee-has-been-a-spectacular-failure>

³⁹ Business studies leadership

⁴⁰ <http://www.redandwhitekop.com/forum/index.php?topic=307914.0>

⁴¹ Mamadou Sakho asked to leave Liverpool admitting that “I turned up late (to a flight and a team meeting) ... There was a rule that I should have observed. I do not at all see myself as above the rules, I respect everyone.”

⁴² <https://www.thisisanfield.com/2017/01/mamadou-sakho-admits-reasons-behind-falling-jurgen-klopp/>

⁴³ Billie Beane get page from previous version

⁴⁴ Beane get page from previous

⁴⁵ Iago Aspas, Sebastian Coates, Javier Manquillo, Raheem Sterling, Rickie Lambert, Rafael Paez, Fabio Borini, Joao Teixeira, Joe Allen, Mario Balotelli, Luis Alberto etc.

⁴⁶ Money ball, Michael Lewis, Chapter 8 on “Scott Hatterberg, Pickin’ Machine”.

⁴⁷ Money ball, Michael Lewis, Chapter 7 on “Giambi’s Hole”.

⁴⁸ Ibid at 46.

H. FINANCE

“Financial distress is part of soccer’s territory”⁴⁹ however FSG has continued to (despite no duty of care)⁵⁰ taken “care of its fans by taking off the debts and converting it into equity”⁵¹ and clearly (as seen particularly in 2015, reinvesting Suarez money) indicating longevity by investing in Anfield to earn future revenues (*see footnote*⁵² on Jean Baptiste’s Say’s Law and correlation between stadium size and revenues).

I. MARKETING AND REPUTATIONAL CAPITAL

The concept of reputational capital in football refers to the “the measures a team is trusted, legitimately in a position of power, premium and recognition, with a stronger willingness among fans to hold support a club in times of crisis.”⁵³ Unfortunately LFC fan trust has declined over the last decade with the champion league track record in 1990-2005 earning enough reputational capital protecting Liverpool as being “stigmatised” as failure but still only a “top four club”. An inability for Klopp to attain champion league football may result in LFC being known as a failure.

J. INTERSECTION BETWEEN TECHNOLOGY AND FOOTBALL

As discussed, the biggest changes FSG bought to LFC were sabermetrics and data, statistics that have been used by customers (fans) for years, but ironically been neglected by football clubs. This technology has extended to include control rooms, analytics, data and digital analytics work to enhance team play and attendance.

3A. LIVERPOOL AS AN ORGANIC ORGANISATION: PICKING THE “TEAM”

A. SEPARATE SCOUTING BOARD

As highlighted at *section 2B*, FSG created a separate selection committee sharing decision making with Klopp, to mitigate the risk of an “incorrect hire” – via an objective rational framework as follows:⁵⁴

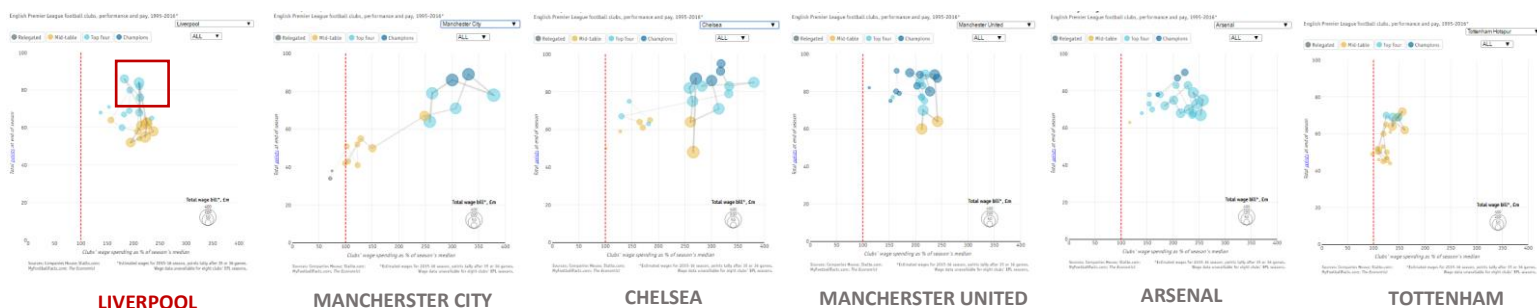
Rules	Decision Making Process
1. Eliminate generalisations of what “typical” is when it is not 2. Accepting that what a player did last was not what they would do first 3. Note that everyone’s single view of reality has an element of illusion which must be validated.	1. Player proposed by individual 2. Process of elimination whereby weaknesses and downfalls debit players from position proposed for (<i>experience + sabermetrics</i>) 3. Base the decision on what remains the key value add (<i>experience + sabermetrics</i>)

Despite being as successful at the Oakland As and Boston Red Sox, LFC has still been troubled with errors (*refer to section 2E*) and as such, Klopp demanded upon joining Liverpool that he would have the “final say” rather than the committee.

B. SPENDING

Szymanski reminds us that teams that paid the highest player salaries over the past 20 seasons won the league 9 times and the club with the next-most-expensive payroll won 7 times. This is depicted below:⁵⁵

Y AXIS (POINTS GAINED), X AXIS (SPEND AS % MEDIAN LEAGUE SPEND), BUBBLE SIZE (TOTAL WAGE IN



The above points v wages as % season median (1995 to 2016) clearly shows that Liverpool’s highest points (light blue top 4 finishes) in fact had smaller bubble sizes (total wage bill in million pounds), whereas all other clubs light blue and dark blue bubbles are substantially larger and far beyond the total all season median wage spend. Such results are explained below.

⁴⁹ Money and Soccer, A soccormomics Guide, Stefan Szymanski Chapters 1 on Dominance and Distress

⁵⁰ Harvard Extension School, MGMT 5515, Richard Volante.

⁵¹ <http://www.dailypost.co.uk/sport/football/football-news/fsg-clearing-69m-liverpool-fcs-10974360>

⁵² There is a +0.75 correlation between stadium size & wages paid (i.e. increased ticket sales, increases revenues that can pay for more “expensive” skilled players), fundamentally based on Jean Baptiste ‘Say’s Law’ that if “you build (demand) fans will come (supply)”, Money and Soccer, A soccormomics Guide, Stefan Szymanski.

⁵³ Reputational Capital, Wikipedia.

⁵⁴ Money ball, Michael Lewis, Chapter 2 on “How to find a ballplayer”.

⁵⁵ <http://www.economist.com/blogs/graphicdetail/2016/05/daily-chart-0>



C. MARGINAL EXPENDITURE

Liverpool's spend is based on "marginal expenditure" and challenges Szymanski's view that "The more you spend on players, the more successful you will on average and in the long run."⁵⁶ However, FSG's investment strategy is to purchase "undervalued" players and "pay less than what the player is worth" using sabermetrics to minimise the cost proponent. This relies on data providing asymmetric information to arbitrage a player's value based on market perception (i.e. portfolio theory). However, by treating the team as a portfolio, LFC has made tactical investments to diversify risk by offsetting risky players purchasing well established and known players (e.g. Milner, Kolo Touré), thereby, building "aggregate team based on an average".⁵⁷ This not only optimises the cost to win, similar to the Oakland A's FSG imposes the "better off test"⁵⁸ which stipulates that any new player should increase LFCs total competitive advantage (i.e. *synergy* must be present)".⁵⁹ This can also result in lack of investment. This is due to sticking to the "minimal viable price" to attain "production line (where cost inefficient players) have been dropped".⁶⁰ For instance, despite young Alberto Moreno and the opportunity to purchase a left back, Klopp identified the ability to transform James Milner into a reliable left back. Some would argue that Milner is a "defective" left back, but as Beane stated "defective players win games"⁶¹ and are cheaper.

D. OVERCOMING POOR BUYING POWER TO ATTRACT HIGH QUALITY PLAYERS

Due to the lack of champions league football and adherence to marginal costs, LFC in recent years has not only lost its buying power, but has experienced competitors outpricing LFC in last minute negotiations (e.g. Tottenham transfer of Dele Alli). Thus, as mentioned, LFC relies heavily on "superior knowledge" (aka information asymmetry) to identify and keep proprietary / discrete information about players that competitors do not see. This "strategic procurement" mindset has often seen LFC refuse to pay what the market "defines as successful" (e.g. LFC refused to offer more than 4,000 pound per week on Dele Alli) based on the theory that market information (e.g. media, scout talk etc.) inaccurately inflates prices and overvalues talent (e.g. £33m Manchester City purchase of Robinho is one of many examples). Thus, Klopp very much followed Szymanski's finding that with buying and growing future superstars - "even if LFC lose, based on an average set of players, fans will come to see the average players become stars" (e.g. Philippe Coutinho etc.). Ironically, this has been part of Klopp's strategy even when at Borussia Dortmund where he grew "unknown" players to superstars (e.g. Lewandowski, Gotze, Reus and Hummels) and is in line with Szymanski's 'magical formula' (that) acquiring top talent, increases win and in turn earning more revenue to buy better players.

E. SELLING PLAYERS AT AN OVERVALUED PRICE

Firstly, upon signing low cost hidden gems, LFC has deliberately stipulated large termination clauses and sell on clauses when that prove lucrative once players become "superstars" and gives LFC bargaining power in renewing the contract or releasing the player for large sums⁶². If the player does not prosper the contract is varied such that it acts as a future and options contract, derivatives (just like stocks and bonds put together), fragments of the player are put together as a "investment bond" and can be recouped by selling the player to a club seeking the AAA+ skills that the player has, knowing that they are also buying CCC- traits that the player does not have.⁶³ Thus making profits off arbitraging the market and now LFC holding the title of "biggest selling club in Europe".⁶⁴ Furthermore, FSG deliberately uses market information to manipulate markets to sell players at premium.⁶⁵ This is premised on the theory that the "league and football globally" is an "ecosystem of agents bidding ..and the hype of the auction"⁶⁶ and dispelling information to high quality/ranked teams within a realm of "privacy between (bidders) and the market value of players", FSG has lured

⁵⁶ Money and Soccer, A soccernomics Guide, Stefan Szymanski Chapters 4 on Revenue.

⁵⁷ Money ball, Michael Lewis, Chapter 7 on "Giambi's Hole".

⁵⁸ Michael porter - <https://hbr.org/1987/05/from-competitive-advantage-to-corporate-strategy>.

⁵⁹ <https://mbsportal.bl.uk/taster/subjareas/busmanhist/mgmtthinkers/porter.aspx>

⁶⁰ Money ball, Michael Lewis, Chapter 7 on "Giambi's Hole".

⁶¹ Money ball, Michael Lewis, Chapter 7 on "Giambi's Hole".

⁶² <http://www.dailymail.co.uk/sport/football/article-3734953/The-lucrative-sell-clauses-John-Stones-netted-Barnsley-7m-Man-City-move.html>

⁶³ Money ball, Michael Lewis, Chapter 10 on "Anatomy of an undervalued pitcher"

⁶⁴ <http://www.mirror.co.uk/sport/liverpool-officially-named-biggest-selling-8881850>

⁶⁵ Money ball, Michael Lewis, Chapter 9 on "The Trading Desk".

⁶⁶ Money ball, Michael Lewis, Chapter 9 on "The Trading Desk".

“emotional sugar daddy”⁶⁷ owners to pay large sums for unwanted Liverpool players (see footnote ⁴⁴). Unfortunately, LFC has previously fallen into the same trap and now are wary of such errors (e.g. Stewart Downing, Andy Carroll etc.).

3B. LIVERPOOL AS AN ORGANIC ORGANISATION: THE PLAYERS (WORKFORCE)

A. CLEARLY DEFINING THE PLAYERS JOB USING METRICS AND KPIS

As highlighted below, LFC operates as one body that is comprised of units (players). However, the player, aka, unit is given clearly defined roles and targets. The science behind this is prioritising very specific target metrics for each position. The again metrics are focused on “marginal or efficiency - production line principle”⁶⁸ detailed in **Section 3C** to minimise the cost per position. This improves the entire team as individual players are incentivised individual players to hit their targets which contribute to the overall success of the team (e.g. Henderson targets are aimed at successful passes - i.e. 2,057 passes more than second place Paul Pogba (27th Feb 17)). Performance is retrospectively validated, nearly on a weekly basis, but is not compared against other players, rather it compares the player to what is expected from themselves in that role. This, unlike the media and fans constantly comparing players “head to head”, focuses players to perform their respective at the level and standard expected to provide a synergy to the entire team. Thus, Szymanski’s win-maximisation is deliberately narrowed to talent maximisation and winning is a bi-product.

B. ONE BODY, NOT A GROUP OF INDIVIDUALS

LFC’s “portfolio” approach, treating the team as one body, emphasises that every unit (player) just like a stock is highly correlated and has a relationship with the entire team / body. Therefore, one player’s failure or success is equivalent to the team’s failure or success. Fundamentally, the players are broken into key traits (e.g. defending, shooting etc.) and sub-skills (Tacking, marking etc.) and when combines as whole should ideally lead to balanced team (*see Appendix 3 for player data*). In 2015-16, LFC has been criticised with its poor defending and this is reflected in the portfolio (excluding goal keeper position) to the right. Also represented are the skills that have position LFC as the team with highest goals (despite no dedicated strikers, with 14 individual players contributing to the goal tally, emphasising that goals must become a bi-product of the team as whole) and having incredible endurance due to an incredibly thin squad. An important call out is that upon the conclusion of the 2014 season LFC saw its leaders such as Steven Gerrard and Jamie Carragher leaving the squad, Klopp’s side, Captain Jordan Henderson “lacking the same level of conviction...as those who went before him”⁶⁹ was criticised as lacking mental leadership. However, the above shows that with the departure of the undisciplined such as Balotelli, Sakho and the purchases of Milner, Klavan and retaining Lucas Klopp has worked to re-build the leadership as seen by the “mentality” metrics. This has been an issue for Klopp as depending on a team of future superstars, misbalancing talent (for young, fluid high pressing fast football) has sacrificed starting positions for natural leaders critiquing Klopp suggesting “Liverpool's struggles against smaller teams may be Jurgen Klopp's style (and selection).”⁷⁰ This is because such high level targets such as maintaining a “high press” or “high speed” etc. do not dispel the individual traits needed and “organisational success are based on disciplined individual statistics”⁷¹ (e.g. a portfolio is only as good as its units).

LFC PORTFOLIO: Moreno, Milner, Matip, Lovren, Clyne, Wijnaldum, Henderson, Mané, Coutinho, Lallana, Sturridge, Roberto Firmino (exc. GK)



C. HOW TO CALCULATE THE COST OF AN ERROR (VOLATILITY OF THE PORTFOLIO)

Furthermore, given the team is an organic organism, the “dynamic state”⁷² despite statistics and paper work, errors and costs will occur, and due to the portfolio approach, one player’s error impacts the entire team. While ad hoc errors are inevitable, ongoing errors must be “mitigated” or removed. Klopp has brought assurance by enhancing internal “pressure”

⁶⁷ Money and Soccer, A soccernomics Guide, Stefan Szymanski Chapters 6 on “Ownership”

⁶⁸ Money ball, Michael Lewis, Chapter 12 on “The Speed of the Idea”.

⁶⁹ <https://www.thefanfieldwrap.com/2017/02/liverpool-lack-leadership-look-captaincy-jordan-henderson/>

⁷⁰ <http://www.espnfc.com.au/club/liverpool/364/blog/post/3059710/liverpool-struggles-against-smaller-teams-may-be-jurgen-klopp-style>

⁷¹ Money ball, Michael Lewis, Chapter 6 on “The Science of Winning an Unfair Game”.

⁷² Money ball, Michael Lewis, Page 134 money ball



from the bench, and rotating players based on meritocracy (e.g. “Simon Mignolet and Loris Karius goalkeeper battle looks like being one that neither will win” but continues to improve both players). FSG’s calculation of costs can be calculated as “cost = before the players’ actions – after player actions”⁷³ and using platonic credits and debits based on player and team metrics (e.g. # of successful tackles – number of unsuccessful tackles). Thereby, each unit is responsible for decreasing their volatility and thereby the team’s volatility. Defenders are held more punishable as an error (that can lead to an opponent’s goal) is a double negative costing titles (e.g. Steven Gerrard’s slip against Chelsea in 2016). That being said, FSG has introduced new technologies such as the Advance Value Matrix⁷⁴ software, to predict “unknowns” (including the opponent) outcomes and has increased given Klopp have internal insights and opponent insights, thus understanding the game as “figures on a chess board.”⁷⁵ However, the “central paradox of soccer is that no matter how much we know about each player, the outcome of every competitive game remains unpredictable.”⁷⁶

D. FINDING HIDDEN GEMS

As mentioned above, LFC’s use of sabermetrics has worked to prove Szymanski’s view that expenditure is a very reliable indicator of team’s success. and invested very early on in data to attain superior information and deep analytics to identify the intrinsic ‘hidden gem’ that many scouts may miss herding towards brand names such as Pogba, Sterling etc. that demand extremely large fees but do not live up to their potential (e.g. the Robam Abramovich is an extremely emotional owner and sugar daddy that has let emotions dictate subjective purchases that without insight and data have led to extreme failures – e.g. Andriy Shevchenko (£30 Million) in 2006 and Fernando Torres in (breaking the English record costing (£50 Million) – with Liverpool using sabermetrics to purchase Luis Suarez for (£22 Million) flourishing at LFC and sold for (£65-75 Million).

E. THE B TEAM AND “JUST IN CASE GUYS”⁷⁷

LFC has created its own “secondary market”. That is the purchase of a second and third squad players consisting of players that scouts would not consider in the premier league. That on paper did not amount to much, but in Klopp’s inorganic organisation do (in terms of second squads etc. and potential substitutes/rotation players (e.g. Jordan Ibe). Such players not only complete Klopp’s picture but earn a revenue for the club, as once a signal of the player “at a big team” a signal / information automatically enters the market and player values are intrinsically valued higher and even extend the years (if maturing) playable (typically purchased in lower leagues). Secondly, Liverpool have purchased “just in case guys” with players knowing they will play the role of sitting on the bench and be used as reliable player when the squad needs relief (including the bench). Such players are opposite of the “finished product”⁷⁸ the Boston red socks have called such players the “AAA guys”. This dynamic is directly addressed at Liverpool like players like Klaven and Kolo Touré accepting contracts to retirement, knowing such roles are to support the team as “just in case guy” and are inspired by novel tasks (e.g. the job of supporting the team, teaching etc.) (see appendix 4: Jeremy Brown) instead of going out of contract and forcing to retire.

F. PHYSICAL ATONOMY AND MENTAL PSYCH OF THE PLAYERS AND THE MANAGER

The physical autonomy of each player feeds into key strategies formed on the pitch (e.g. Matip is deliberately used as a target on corners due to his height) and can be a key decider in whether the player fits the team (e.g. Andy Carroll physical way of playing contradicted Klopp’s fluid way of playing). This is decided in the scouting department as part of the process of elimination highlighted **section 3AA**. Mentally, however, LFC has imposed that players are to be treated as a normal workforce and as “humans”. Thus, considerations such as mental distractions, emotions, is managed as Beane put it as the “art” in sports and is considered not only at the team level but at the individual player level. This is as mentioned, is

⁷³ Money ball, Michael Lewis, Chapter 6 on “The Science of Winning an Unfair Game”.

⁷⁴ <http://grantland.com/features/2015-mlb-avm-systems-ken-mauriello-jack-armbruster-moneyball-sabermetrics/>

⁷⁵ Money ball, Michael Lewis, Page 213.

⁷⁶ Money and Soccer, A soccernomics Guide, Stefan Szymanski Chapters 1 on “Dominance and Distress”

⁷⁷ Money and Soccer, A soccernomics Guide, Stefan Szymanski

⁷⁸ Science and Soccer: Developing Elite Performers, A. Mark Williams, 3rd Edition, Routledge.



due to players being whole with the team with success resting on the units that build the “portfolio”. Such emphasise is placed as the psychology of the players is also correlated with the strength of the squad⁷⁹ and is what creates the “law of competitive balance” whereby players are encouraged to focus on weaknesses rather than strengths and that winning is not the target. This is done to moderate the superstar and prevent overconfidence that can significantly hinder the team’s performance (e.g. in the 2016-17 LFC notable struggled to beat lower league teams). However, players are naturally “superstars” from day one, but in addition to over confidence, players have history of habit, routine and errors embedded in their DNA. LFC has approached particular player habits, by respecting them and then exploiting⁸⁰ them for the team’s benefit but players who are unable to change mentally to new routines or become a risk to the team are “monitored and traded when psychologically out of control⁸¹ (e.g. Mario Balotelli had agreed to a number of agreements to control his psyche in order to sign at LFC). This assessment extended beyond the pitch and statistical driver are ran in computerised profiles (e.g. 1-10 leadership, 1-10 ambition)⁸² in order to understand alternative influencers (e.g. alcoholism, family situation etc.) and potentially identify and mitigate “flight risk” to competitors (e.g. Raheem Sterling, Fernando Torres etc.).⁸³

Likewise, even managers are expected to make mental adjustments. Klopp has used the same principles of Darwinism’s to adapt. Klopp enforced his control on his first day stating “we have to do it one way and that is my way you do not have much choice.”⁸⁴ Players and fans reacted positively as Klopp reinforced that a key ingredient to bringing back the glory days was “11 players that want to help the team. That to listen to what I say...(as it) is to have 11 players doing the same thing wrong than everybody doing what they want...Then it is more probable that you win and become successful.”⁸⁵ However, the over focusing on individual development in order for the success of the team must be reminded as players do react with burnout and often lack of purpose sitting on the bench or over training and burning out (as recently players stated depression⁸⁶ due to Klopp’s harsh training requirements (e.g. training at night etc.), **See appendix 5** for leaked training schedule). Thus, the manager must remember that focus on the player(s) bottom up then the whole team and but reinforce without the players there can be no team and never forget morale.

⁷⁹ Money and Soccer, A soccernomics Guide, Stefan Szymanski Chapters 11 on “The Human Element”

⁸⁰ Money ball, Michael Lewis, Page 260.

⁸¹ Money ball, Michael Lewis, Page 257.

⁸² Money ball, Michael Lewis, Chapter 2 on “How to find a ballplayer”.

⁸³ Money ball, Michael Lewis, Chapter 2 on “How to find a ballplayer”.

⁸⁴ <https://www.theguardian.com/football/blog/2015/nov/09/liverpool-fans-jurgen-klopp-belief-optimism>

⁸⁵ <http://www.telegraph.co.uk/sport/football/teams/liverpool/11923605/Jurgen-Klopp-warns-We-will-do-things-one-way-at-Liverpool-and-that-is-my-way.html>

⁸⁶ <http://www.dailystar.co.uk/sport/football/football-news/liverpool-stars-left-depressed-jurgen-9925096>

4. CONCLUSION AND RECOMMENDATIONS

Thus, overall how has Liverpool achieved its success? From the above we draw clarity and can offer recommendations:

- The Operating Model of separating the inorganic team and executive team can only succeed if the interlink (manager – Jurgen Klopp) does not maintain a balanced view of being a “manager” but also “equal to the players” and therefore not getting lost in back office activities
- Middle managers and others not involved in the direct lines of decision making should have a voice. It is likely that the players and managers, coaches and fans voices are heard as it is the “inorganic” entity that forms the heart of any football club
- John Henry proves to be a successful pioneer in turning clubs around and revolutionising sports teams however, the FSG must be careful and not attempt to impose past success from other sports, games, experiences forcing Liverpool to take a route that will damage its progress thus far.
- Jurgen Klopp as the Manager must maintain his view as “the normal one” as losing the trust of the of the policy makers or players or fans will result in miscoordination of LFC and FSG strategic plans, and interruption of long term success will be disrupted by a new manager.
- Players are treated as a normal “workforce” however, this should not be over exaggerated in a way that deteriorates the players sense of purpose for the game. For instance, players playing out of position or as substitutes as plan B players can result in an uneasy culture in the locker room and ultimately ultimatums and demands that does not FSG and the players long term goals.
- The corporate culture should accommodate all as equals, but as understood as many players, superstars need to be introduced into the clubs, implying higher wages, status etc. and all players should be educated on the fact that such stars are improving the overall standard, revenue and attendance of fans to LFC games.
- Campaigns should be introduced immediately in order to protect LFC reputational capital as a further slip in not securing consistent champions league positions will highly hinder fan support.
- Correctly, the scouting committee should take Jurgen Klopp’s recommendations and advice as crucial as it is the players that feed into his on-field plan that leads to the success of LFC as corporation.
- As soon as LFC attain champions league consistency it should invest in highly skilled players in order to maintain the skill but also draw in fans and revenues in search of the “golden days”.
- Klopp needs to invest in defenders, this is clear by the analysis in 3B.
- Hidden gems should be continuing to be identified, but the strictness of maximum spend should be adhered to as due to the organic nature of football, statistics such as sabermetrics do not promise future success. Likewise, contracting techniques to re-coup value should continue – if not become more stringent if LFC does find itself as a recognised top four club again (increased bargaining power).
- The psychology of the team should be maintained, but over oppressive conditions should constrain lifestyles outside of football such tat players are “depressed” or no longer enjoy life, and as an impact will no longer enjoy playing football.
- “You Never Walk Alone”.



APPENDIX 1: DIALECTICS APPLIED TO SOCCER

Dialectics is a reasonable philosophy to justify Jurgen Klopp's view that errors are necessary decision and moments that lead to positive outcomes.

Dialectic or dialectics (Greek: διαλεκτική, dialektikḗ), also known as the dialectical method, is a discourse between two or more people holding different points of view about a subject but wishing to establish the truth through reasoned arguments. Hegelian dialectics is based upon four concepts:

1. Everything is transient and finite, existing in the medium of time.
2. Everything is composed of contradictions (opposing forces).
3. Gradual changes lead to crises, turning points when one force overcomes its opponent force (quantitative change leads to qualitative change).
4. Change is helical (periodic without returning to the same position), not circular (negation of the negation).

<https://en.wikipedia.org/wiki/Dialectic>

APPENDIX 2: STURRDIGE ON PITCH NOTES

- Liverpool manager gave striker piece of paper during EFL Cup tie
- Forward ran around offering instructions to other players



① Daniel Sturridge reads a piece of paper given to him by Jürgen Klopp as he runs around during the EFL Cup semi-final against Southampton. Photograph: James Marsh/BPI/Hex/Shutterstock

<https://www.theguardian.com/football/2017/jan/11/liverpool-football-jurgen-klopp-daniel-sturridge-note-efl-cup-southampton>.

APPENDIX 3: LFC STARTING 11 RATINGS

LFC																													
Pos	Player	Progra	OVR	Pace					Shot				Pass				Tech		Head	Deff		Phys				Ment			
				AC	SP	FI	SH	LS	FR	CR	SP	LP	BA	DF	HE	MR	TA	ST	ST	AG	PQ	AW	RE	CO					
LWB	Ibeto Moren	Basic	75	89	89	65	80	74	40	79	74	64	81	77	61	67	74	63	85	83	73	70	68	57					
LB	Ibeto Moren	Basic	74	89	89	65	80	74	40	79	74	64	81	77	61	67	74	63	85	83	73	70	68	57					
LB	Milner	OOP	80	75	72	74	93	87	85	92	91	88	84	83	75	73	74	77	97	77	83	93	90	92					
CB	Matip	BD	90	75	85	44	74	56	30	62	85	85	79	76	96	93	92	87	83	78	59	92	89	87					
CB	Matip	Basic	82	67	77	36	66	48	22	54	77	77	71	68	88	85	84	79	75	70	51	84	81	79					
CB	Sakho	Basic	80	55	68	23	51	38	30	60	75	66	61	50	71	79	84	90	72	85	32	70	72	73					
CB	Lovren	Basic	81	61	55	33	63	34	18	59	70	68	67	64	75	82	85	85	70	87	21	79	75	75					
RB	Clyne	NC	79	86	86	54	74	69	39	80	76	67	76	78	66	80	83	71	87	76	76	67	75	75					
RB	Clyne	Basic	79	86	86	54	74	69	39	80	76	67	76	78	66	80	83	71	87	76	72	67	75	75					
RWB	Clyne	Basic	80	86	86	54	74	69	39	80	76	67	76	78	66	80	83	71	87	76	72	67	75	75					
LM	Wijnaldum	BD	89	88	94	85	86	78	80	76	88	75	93	95	81	61	64	73	95	70	87	92	91	87					
LM	Wijnaldum	TT	82	79	90	80	77	69	71	71	81	66	87	86	72	52	55	64	86	61	82	88	82	78					
LM	Wijnaldum	Basic	80	79	85	76	77	69	71	67	79	66	84	86	72	52	55	64	86	61	78	83	82	78					
CM	Henderson	TH	93	92	89	82	90	86	86	96	98	96	93	89	80	83	90	89	108	93	89	98	94	93					
CM	Milner	SC	86	69	75	76	87	86	79	86	90	88	85	83	69	75	75	71	91	71	82	92	89	86					
CM	Milner	Basic	80	69	66	68	87	81	79	86	85	82	78	77	69	67	68	71	91	71	77	87	84	86					
RM	Mané	AM	94	109	107	94	91	84	82	86	91	85	95	99	71	51	51	90	83	79	95	69	93	92					
RM	Mané	Basic	76	93	91	76	73	66	64	68	73	67	77	84	53	33	33	72	65	61	77	51	75	74					
CAM	Coutinho	CNBLM	94	100	92	85	92	93	87	87	98	91	100	100	60	37	51	69	86	48	89	86	88	92					
CAM	Coutinho	CNBL	89	95	87	80	87	88	82	82	93	86	95	95	55	32	46	64	81	43	84	81	83	87					
CAM	Lallana	BD	91	89	83	86	81	85	79	87	96	87	95	98	80	43	65	80	96	59	89	88	88	90					
CAM	Coutinho	MOTM	88	96	84	77	86	87	81	81	92	85	95	94	54	31	45	63	80	42	83	80	82	86					
CAM	Coutinho	NC	83	89	77	70	79	85	72	74	90	78	88	92	47	24	38	56	73	35	80	73	75	79					
CAM	Coutinho	Basic	81	89	77	70	79	80	74	74	85	78	88	87	47	24	38	56	73	35	76	73	75	79					
LF	Sturridge	Basic	82	88	89	85	83	83	68	62	75	62	81	84	73	23	24	73	72	60	84	79	82	81					
CF	Ibeto Firmii	AM	95	93	95	96	93	93	81	84	96	88	97	101	85	38	66	85	93	78	95	85	94	90					
CF	Balotelli	CNBL	85	86	88	91	86	82	88	78	80	62	95	91	72	31	32	95	65	79	85	70	70	74					
CF	Ibeto Firmii	IF	83	81	81	84	81	82	67	70	84	74	84	89	73	24	52	71	79	64	83	73	81	76					
CF	Ibeto Firmii	Basic	81	79	79	82	79	79	67	70	82	74	83	87	71	24	52	71	79	64	81	71	80	76					
RW	Mané	IP	78	94	92	77	74	67	65	69	74	68	78	85	54	34	34	73	66	62	78	52	76	75					
RW	Mané	Basic	77	93	91	76	73	66	64	68	73	67	77	84	53	33	33	72	65	61	77	51	75	74					
RW	Marković	Basic	77	91	92	67	65	60	62	67	73	66	82	84	58	30	39	56	71	41	74	70	74	80					
ST	Ibeto Firmii	BD	93	92	92	95	92	92	80	83	95	87	96	100	84	37	65	84	92	77	94	84	93	89					
ST	Sturridge	TH	88	88	92	90	89	92	68	62	81	62	81	84	85	23	24	82	84	60	93	79	91	84					
ST	Sturridge	Basic	82	88	89	85	83	83	68	62	75	62	81	84	73	23	24	73	72	60	84	79	82	81					

Original Source: Futhead 2016-17.

APPENDIX 4: Jeremy Brown Oakland As

The Oakland as kept Jeremy Brown in their squad paying him \$350k (almost less than ~\$1 million expected to be paid for a 35-draft pick) as he enjoyed the novel task of playing while losing weight. Knowing this information Brown was a particularly cheap buy given he was bought to “walk” and show “power” and his value indeed increased overtime.

APPENDIX 5: Liverpool's training regiment

12-day regiment based on 3 matches:

Sat	Sun	Mon	Tues	Wed	Thur	Fri	Sat	Sun	Mon	Tues	Wed
3pm - game	11am - recovery session	day off	10:30am - light training / 5pm - heavy training	5pm - training	3:30pm - training	5pm - training	5:30pm - game	11am - recovery session	5pm - training	5pm - training	10:30am - training / 8pm game

<http://www.dailystandard.co.uk/sport/football/football-news/liverpool-stars-left-depressed-jurgen-9925096>